

Autodesk® **Scaleform®**

Scaleform 4.4

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Autodesk® Scaleform® 4.4

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1 はじめに

Scaleform 4.0

Scaleform SDK

Scaleform 4.0 API

Scaleform

Scaleform 4.4

API

3.3

2 マルチスレッド型レンダラーへの変更

Scaleform 3.x

Scaleform 4.0

API

1. レンダラーの生成

2. レンダラー キャッシュの構成

GFxLoader

3. テクスチャ リソースの生成

4. レンダリング ループ

MovieDisplayHandle

HAL::Display

5. ムービーのシャットダウン GFx::Movie

6. カスタム レンダラー

GRenderer

API

Render::HAL

Scaleform 3.x

2.1 レンダラーの生成

Scaleform
 Render::HAL
 Display
 Render::D3D9::HAL Render::PS4::HAL

```
D3DPRESENT_PARAMETERS    PresentParams;
IDirect3DDevice9*        pDevice = ...;
ThreadId                 renderThreadId = Scaleform::GetCurrentThreadId();
Render::ThreadCommandQueue *commandQueue = ...;
Ptr<Render::D3D9::HAL>    pHal;

pHal = *SF_NEW Render::D3D9::HAL(commandQueue)
if (!pHal-> InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
renderThreadId)))
{
    return false;
}
```

HALInitParams
 Scaleform
 HALInitParams D3D9
 pDevice PresentParams
 Platform HALInitParams
 renderThreadId
 ThreadCommandQueue
 Scaleform
 HAL
 HAL

TextureManager RenderBufferManager MemoryManager HALInitParams

MemoryManager Scaleform
PS3 PS4 PSVita

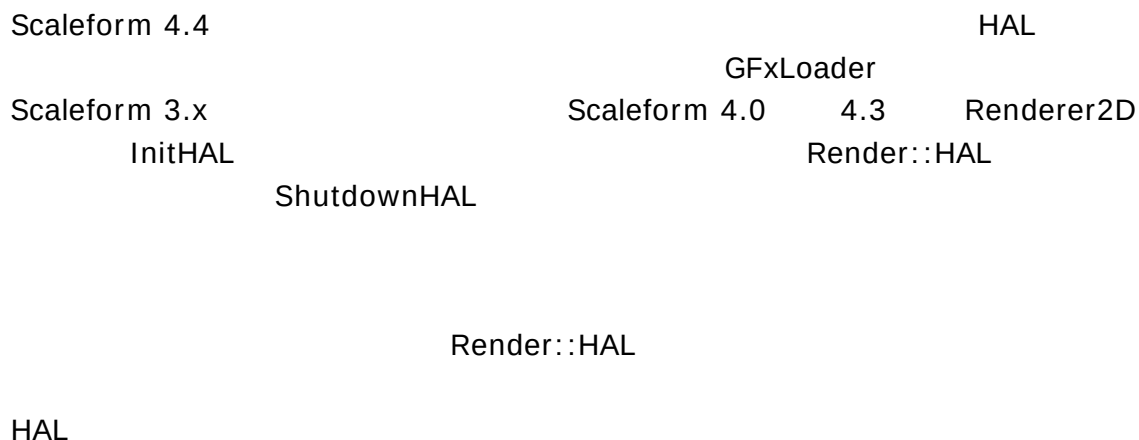
HALInitParam 'halConfigParams'

フラグ	詳細
Render::HALConfig_SoftwareDeferredContext	Scaleform 4.4 D3D9 + GL : HAL::Submit
D3D9::HALConfig_NoSceneCalls	IDirect3DDevice9::BeginScene IDirect3DDevice9::EndScene Scaleform
D3D9::HALConfig_StaticBuffers	Scaleform
D3D9::HALConfig_ShaderModel20	Scaleform D3D9 Shader Model 3.0 Shader Model 2.0
GL::HALConfig_DynamicShaderCompile	InitHAL

GL::HALConfig_DisableBinaryShaders	
GL::HALConfig_MultipleShaderCacheFiles	HALConfig_DynamicShaderCompile
GL::HALConfig_DisableShaderPipelines	Separate Separate
GL::HALConfig_DisableMultipassShaderCompile	Scaleform

~ 4 =

2.2 レンダー キャッシュの設定



HAL

```

MeshCacheParams mcp;
mcp.MemReserve      = 1024 * 1024 * 3;
mcp.MemLimit        = 1024 * 1024 * 12;
mcp.MemGranularity  = 1024 * 1024 * 3;
mcp.LRUTailSize     = 1024 * 1024 * 4;
mcp.StagingBufferSize = 1024 * 1024 * 2;

pHAL->->SetMeshCacheParams(mcp);

```



```

GlyphCacheParams gcp;
gcp.TextureWidth  = 1024;
gcp.TextureHeight = 1024;

```

```

gcp.NumTextures    = 1;
gcp.MaxSlotHeight = 48;

pHAL->->SetGlyphCacheParams(gcp);

```

```

TextureWidth  TextureHeight      NumTextures
              MaxSlotHeight

```

```

              InitHAL
InitHAL      SetParams

```

2.3 テクスチャ リソースの生成

```

Scaleform 4.x      Gfx::ImageCreator
                  Render::TextureManager
                  ImageCreator  Gfx::Loader
                                Scaleform

```

```

Gfx::Loader loader;
...

SF::Ptr<Gfx::ImageCreator> pimageCreator =
    *new Gfx::ImageCreator(pRenderThread->GetTextureManager());
loader.SetImageCreator(pimageCreator);

```

```

ImageCreator      TextureManager

Render::HAL::GetTextureManager()  TextureManager
                                   GetTextureManager

```

CreateTexture

TextureManager

Render::HAL

-

D3DCREATE_MULTITHREADED

D3D9

-

Render::HAL

ThreadCommandQueue

D3D デバイスを使ってマルチスレッド レンダリングを行うには、テクスチャの生成に対して、ThreadCommandQueue の実装と必要に応じてそのサービスが必要

Platform::RenderHALThread

2.4 レンダリング ループ



1. **初期化** – `Gfx::MovieDef::CreateInstance`

`Gfx::Movie`

2. **メイン スレッド処理ループ** –

Advance

ActionScript

Advance

Invoke/DirectAccess API

3. レンダリング –

MovieDisplayHandle

```

//-----
// 1.

Ptr<Gfx::Movie>      pMovie = ...;
Gfx::MovieDisplayHandle hMovieDisplay;

//
//
pMovie->SetViewport(width, height, 0,0, width, height);
hMovieDisplay = pMovie->GetDisplayHandle();

//
// Scaleform Player
pRenderThread->AddDisplayHandle(hMovieDisplay);

//-----
// 2.

//                               ExternalInterface
//                               Advance
float deltaT = ...;
pMovie->HandleEvent(...);
pMovie->Advance(deltaT);

//
//
pRenderThread->WaitForOutstandingDrawFrame();
pRenderThread->DrawFrame();

//-----
// 3.           -           DrawFrame
Ptr<Render::HAL> pHAL = ...;

pHAL->BeginFrame();
bool hasData = hMovieDisplay.NextCapture(pHAL->GetContextNotify());
if (hasData)
    pHAL->Display(hMovieDisplay);
pHAL->EndFrame();

```

- WaitForOutstandingFrame – CPU 2

Advance

- MovieDisplay.NextCapture – Advance

false

2.5 ムービーのシャットダウン

GFX::Movie Release

Render::ThreadCommandQueue

ThreadCommandQueue

Movie::ShutdownRendering(false)

Movie::IsShutdownRenderingComplete

NextCapture

false

3 マルチスレッド型レンダリングのコンセプト

Scaleform
レッド

アドバンス スレッド レンダリング ス

GFX::Movie::Advance

ActionScript

Scaleform

Render::HAL

Scaleform

-

-

Scaleform Player

Scaleform

3.1 ムービーのスナップショット

Advance

Scaleform

Gfx::Movie::Advance
Gfx::Movie::Capture

NextCapture

- Movie::Advance

- Capture NextCapture

Capture
Capture

- Invoke Direct Access API
Capture Advance

Scaleform 4.0

3.3 Movie::Invoke

Display
Capture Advance
Invoke

Scaleform

Advance

3.2 レンダリング状態の保存

Render::HAL::Display

Render::HAL::Display

Display
Scaleform

3.3 マルチコア レンダリング

Scaleform 4.4

CPU

API

API

Scaleform

プラットフォーム	詳細
D3D9	HALInitParams ConfigFrag HALConfig_SoftwareDeferredContext
D3D11	D3D1x::HALInitParams pD3DContext ID3D11Device::CreateDeferredContext ()
XBox One	D3D1x::HALInitParams pD3DContext ID3D11Device::CreateDeferredContext ()
Playstation 4	
WiiU	4.4
OpenGL/GLES	HALInitParams ConfigFrag HALConfig_SoftwareDeferredContext

HAL::Submit
HAL::BeginScene/HAL::EndScene
HAL::EndScene
HAL::Submit
Scaleform

PlayStation 4

HAL::Submit

PS4::HAL::AppendCommandBuffersToList

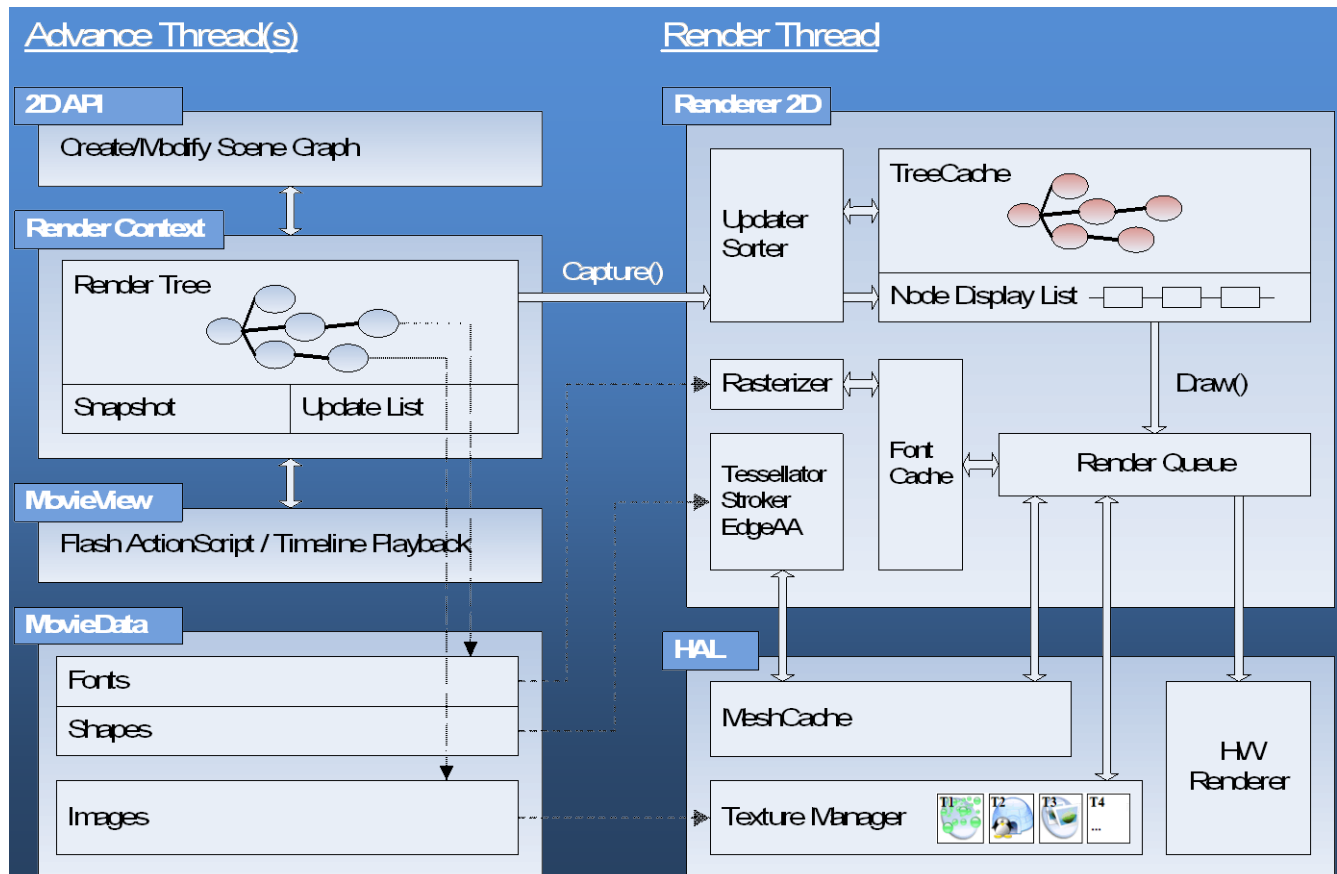
sce::Gnm::submitCommandBuffers

Scaleform

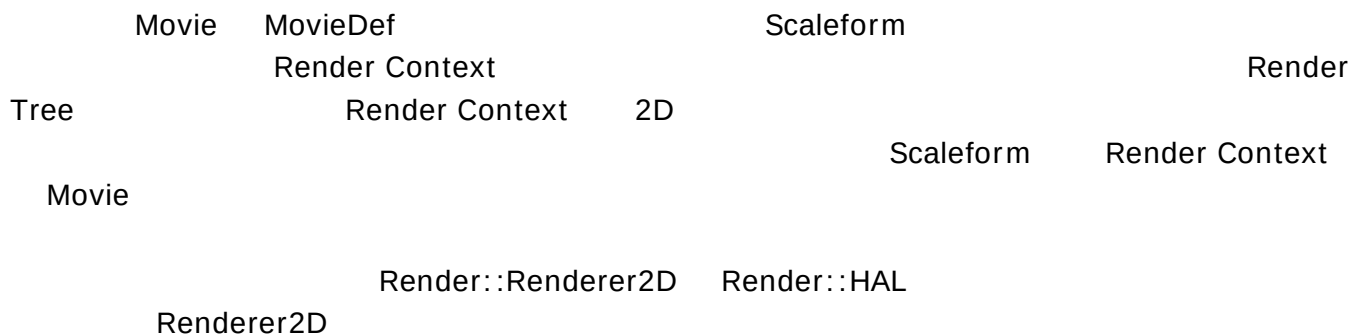
PS4::HAL::SetGfxContext

4 レンダラーのアーキテクチャ

Scaleform 4.0



1.



Display

Render::HAL HAL

Render HAL

-

- Font Cache

Font Cache

-

- Mesh Cache

- Render Queue
CPU

/

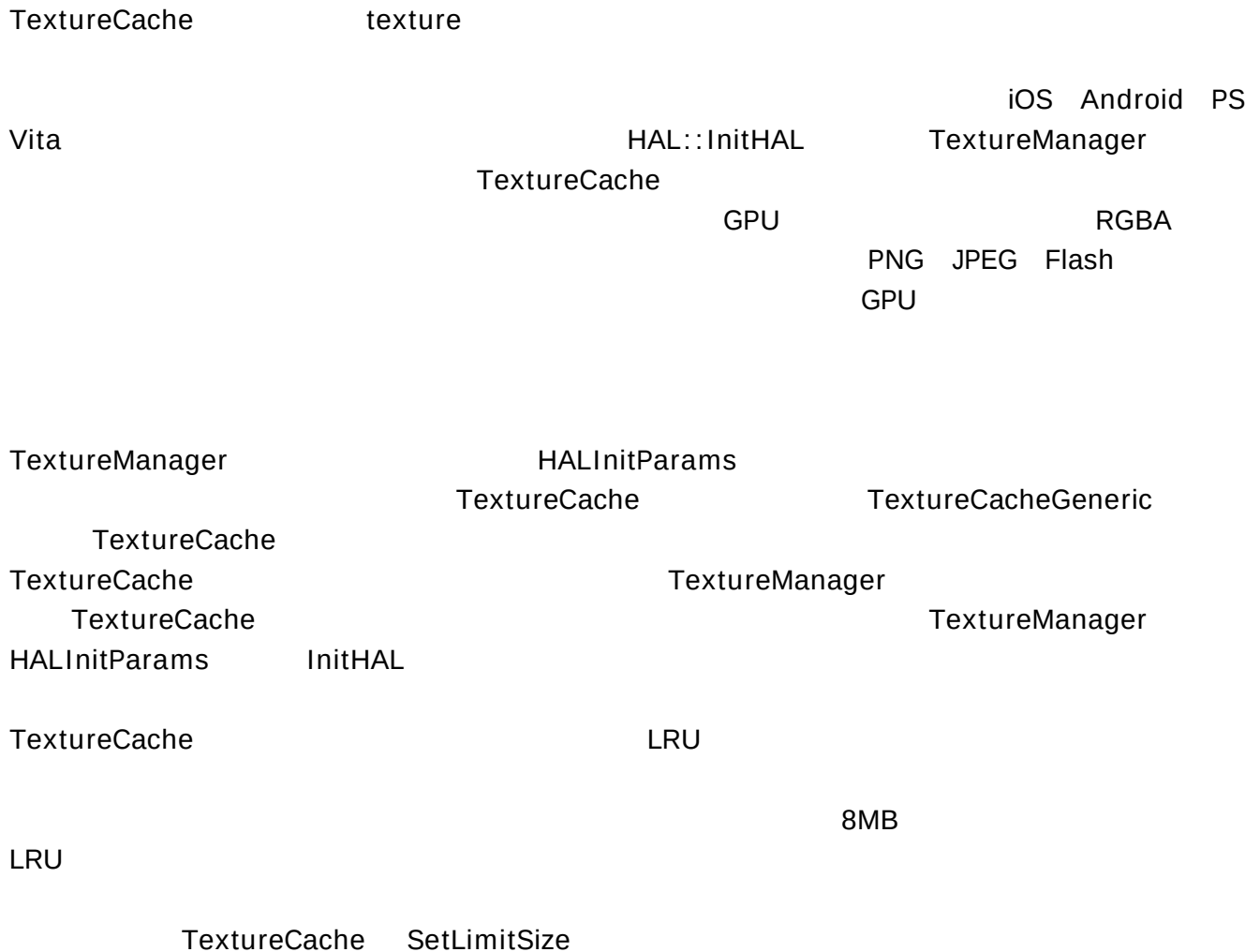
- HAL API

ActionScript

AI

5 サブシステム記述子のレンダー

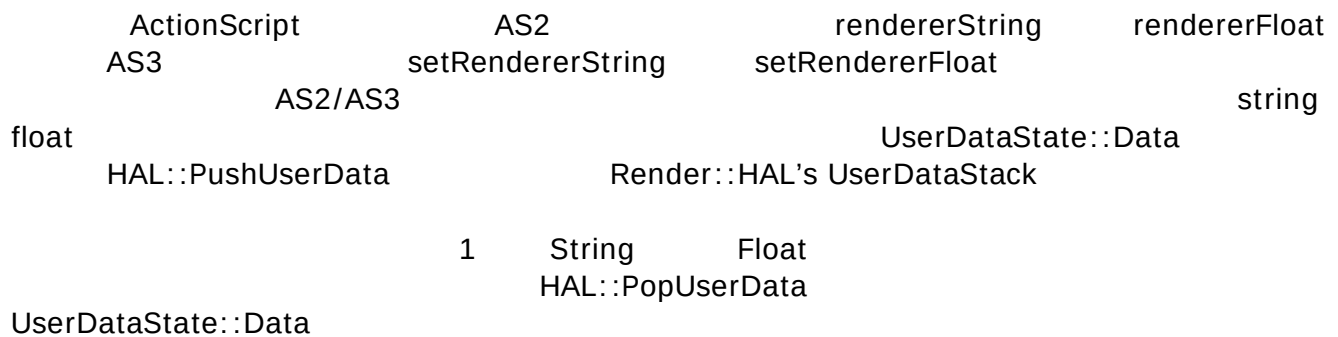
5.1 TextureCache



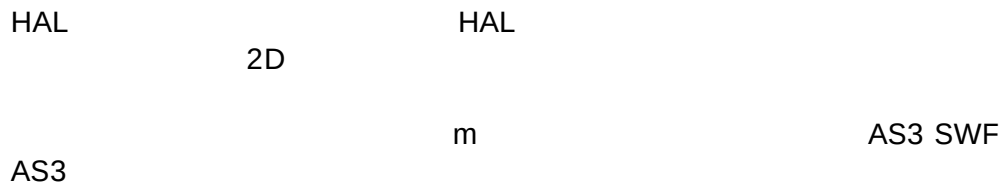
6 カスタムデータのレンダリング

Scaleform

6.1 形状レンダリングの阻止



6.2 カスタムレンダリングの例



```
import scaleform.gfx.*;
DisplayObjectEx.setRendererString(m, "Abc");
DisplayObjectEx.setRendererFloat(m, 17.1717);
```

4.4

Src/Render/ShaderData.xml

```
<ShaderFeature id="Position">
  <ShaderFeatureFlavor id="Position2d" hide="true"/>
  <ShaderFeatureFlavor id="Position3d"/>
  <ShaderFeatureFlavor id="Position2dOffset"/>
</ShaderFeature>
```

Position2dOffset
Position2dOffset

```
<ShaderSource id="Position2dOffset" pipeline="Vertex">
    Position2d(
    attribute float4 pos      : POSITION,
    varying   float4 vpos    : POSITION,
    uniform   float4 mvp[2],
    uniform   float  poffset)
    {
    vpos = float4(0,0,0,1);
    vpos.x = dot(pos, mvp[0]) + poffset;
    vpos.y = dot(pos, mvp[1]) + poffset;
    }
</ShaderSource>
```

Position2d x y poffset

D3D9_Shader.cpp

ShaderInterface::SetStaticShader:

```
// See if we have user data on the stack.
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
    data->RendererString.CompareNoCase("abc") == 0)
    {
        // Modify the shader we use, if we find the matching user data we were
expecting.
        vshader = (VertexShaderDesc::ShaderType)((int)vshader +
VertexShaderDesc::VS_base_Position2dOffset);
        shader  = (FragShaderDesc::ShaderType) ((int)shader +
FragShaderDesc::FS_base_Position2dOffset);
    }
}
```

HAL UserDataStack
abc

Position2dOffset

ShaderData.xml

D3D9_ShaderDescs.h

poffset
ShaderInterface::Finish:

```
// See if we have user data on the stack.
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
        data->RendererString.CompareNoCase("abc") == 0)
    {
        // Add the poffset uniform into the uniform data.
        for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
        {
            float poffset = data->RendererFloat / 256.0f;
            SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
        }
    }
}
```

meshCount

meshCount
HAL

m

6.3 カスタムドロローのバッチを無効にする

HAL Scaleform

Scaleform

Scaleform

AS2 “disableBatching”

AS3

“disableBatching”

AS2/AS3

7 GFxShaderMaker

7.1 概要

Scaleform 4.1 GFxShaderMaker Scaleform
Render_D3D9
ShaderData.xml

D3D9

Src/Render/D3D9_ShaderDescs.h
Src/Render/D3D9_ShaderDescs.cpp
Src/Render/D3D9_ShaderBinary.cpp

7.2 複数シェーダーAPIをサポートする HAL

7.2.1 D3D9

D3D9 HAL ShaderModel 2.0 ShaderModel 3.0
HAL

GFxShaderMaker '-shadermodel'

GFxShaderMaker.exe -platform D3D1x -shadermodel SM30

ShaderModel 2.0 ShaderModel 2.0
InitHAL

7.2.2D3D1x

D3D1x HAL

3

D3D11 D3D10.1

FEATURE_LEVEL_10_0

FEATURE_LEVEL_9_3

FEATURE_LEVEL_9_1

HAL

GfxShaderMaker '-featurelevel'

GfxShaderMaker.exe -platform D3D1x -featurelevel

FEATURE_LEVEL_10_0, FEATURE_LEVEL_9_1

FEATURE_LEVEL9_3

FEATURE_LEVEL_9_3

InitHAL

FEATURE_LEVEL_9_1